

HW3

Lucy Gagnon

Question 2

The Uniform-Pareto

The goal of this problem is to continue getting more practice calculating the posterior distribution.

Suppose $a < x < b$. Consider the notation $I_{(a,b)}(x)$, where I denotes the indicator function. We define $I_{(a,b)}(x)$ to be the following:

$$I_{(a,b)}(x) = \begin{cases} 1 & \text{if } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

Let X be a random variable and let x be an observed value. Let

$$\begin{aligned} X = x \mid \theta &\sim \text{Uniform}(0, \theta) \\ \theta &\sim \text{Pareto}(\alpha, \beta), \end{aligned}$$

where $p(\theta) = \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}} I_{(\beta,\infty)}(\theta)$. Write out the likelihood $p(X = x \mid \theta)$.

Then calculate the posterior distribution of $\theta \mid X = x$.

Hint 1: To set up this problem, do the following:

$$p(\theta \mid x) \propto \frac{1}{\theta} I_{(0,\theta)}(x) \times \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}} I_{(\beta,\infty)}(\theta).$$

Hint 2: You cannot drop the indicators. You must write the indicators as functions of θ since this is the random variable.

Hint 3: It will end up being an updated Pareto.

Proof. Consider the notation $I_{(a,b)}(x)$, where I denotes the indicator function. We define $I_{(a,b)}(x)$ to be the following:

$$I_{(a,b)}(x) = \begin{cases} 1 & \text{if } a < x < b, \\ 0 & \text{otherwise.} \end{cases}$$

Let X be a random variable and let x be an observed value. Let

$$\begin{aligned} X = x \mid \theta &\sim \text{Uniform}(0, \theta) \\ \theta &\sim \text{Pareto}(\alpha, \beta), \end{aligned}$$

where $p(\theta) = \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}}I_{(\beta,\infty)}(\theta)$. The likelihood $p(X = x \mid \theta)$ may be written as

$$p(X = x \mid \theta) = \left(\frac{1}{\theta - 0}\right) * \mathbf{I}_{(0,\theta)}(x) = \left(\frac{1}{\theta}\right) * \mathbf{I}_{(0,\theta)}(x).$$

We will calculate the posterior distribution of $\theta \mid X = x$. By Bayes' Theorem, observe

$$p(\theta \mid x) = \frac{p(x, \theta)}{p(x)} = \frac{p(x \mid \theta) * p(\theta)}{p(x)} \propto p(x \mid \theta) * p(\theta).$$

when we absorb terms that do not depend on θ (i.e., $p(x)$) into the constant of proportionality. Substituting in our likelihood and prior gives

$$\begin{aligned} p(x \mid \theta) * p(\theta) &= \left(\left(\frac{1}{\theta}\right) * \mathbf{I}_{(0,\theta)}(x)\right) * \left(\frac{\alpha\beta^\alpha}{\theta^{\alpha+1}}I_{(\beta,\infty)}(\theta)\right) \\ &\propto \mathbf{I}_{(0,\theta)}(x) \frac{1}{\theta^{(\alpha+1)+1}}I_{(\beta,\infty)}(\theta). \end{aligned}$$

When we absorb terms not depending on θ into the constant of proportionality. To simplify the indicator functions into 1 term, we must consider the region where both

$$\mathbf{I}_{(0,\theta)}(x) = 1$$

and

$$\mathbf{I}_{(\beta,\infty)}(\theta) = 1,$$

which is true for $x < \theta$. Thus, we have

$$\mathbf{I}_{(0,\theta)}(x) \frac{1}{\theta^{(\alpha+1)+1}} I_{(\beta,\infty)}(\theta) = \frac{1}{\theta^{(\alpha+1)+1}} I_{(\max(\beta,x),\infty)}(\theta) \propto \text{Pareto}(\alpha' = \alpha + 1, \beta' = \max(\beta, x)).$$

Therefore, the posterior distribution of $\theta|X = x$ is an updated Pareto distribution,

$$\theta | X = x \sim \text{Pareto}(\alpha' = \alpha + 1, \beta' = \max(\beta, x)).$$

Question 3

The Bayes estimator or Bayes rule

The goal of this problem is to practice a similar problem that we considered in Module 2, where we derived the Bayes rule under squared error loss and found the result was the posterior mean.

Part A

Find the Bayes estimator (or Bayes rule) when the loss function is $L(\theta, \delta(x)) = c (\theta - \delta(x))^2$, where $c > 0$ is a constant.

Proof. Suppose we have the loss function $L(\theta, \delta(x)) = c (\theta - \delta(x))^2$, where $c > 0$ is a constant. We will find the Bayes estimator, $\hat{\delta}(x)$. Observe

$$\begin{aligned} \rho(\theta, \delta(x)) &= E[L(\theta, \delta(x)) | X] \\ &= E[c(\theta - \delta(x))^2 | X] \\ &= E[c(\theta^2 - 2\theta\delta(x) - \delta(x)^2) | X] \\ &= c(E[\theta^2 | X] - 2\delta(x)E[\theta | X] + \delta(x)^2). \end{aligned}$$

To find the Bayes estimator, we take the derivative with respect to $\delta(x)$

$$\frac{\partial \rho(\theta, \delta(x))}{\partial \delta(x)} = c(-2E[\theta | X] + 2\delta(x))$$

and solve for $\hat{\delta}(x)$ when $\frac{\partial \rho(\theta, \delta(x))}{\partial \delta(x)} = 0$. Observe

$$0 = c(-2E[\theta | X] + 2\hat{\delta}(x)).$$

Solving for $\hat{\delta}(x)$ gives $\hat{\delta}(x) = E[\theta | X]$. To check if the solution is unique, it must be true that $\frac{\partial^2 \rho(\theta, \delta(x))}{\partial \delta(x)^2} > 0$. Here, we find

$$\frac{\partial^2 \rho(\theta, \delta(x))}{\partial \delta(x)^2} = 2c > 0.$$

since $c > 0$. Thus, our Bayes estimator is unique. QED

Part B

Derive the Bayes estimator (or Bayes rule) when $L(\theta, \delta(x)) = w(\theta)(g(\theta) - \delta(x))^2$. Assume that $w(\theta) > 0$.

Hint: Do so without writing any integrals. Note that you can write $\rho(\theta, \delta(x)) = E[L(\theta, \delta(x)) | X]$. Don't forget to prove or state why the Bayes rule(s) are unique.

Proof. Suppose we have the loss function $L(\theta, \delta(x)) = w(\theta)(g(\theta) - \delta(x))^2$, where $w(\theta) > 0$ is a constant. We will find the Bayes estimator, $\hat{\delta}(x)$. Observe

$$\begin{aligned} \rho(\theta, \delta(x)) &= E[L(\theta, \delta(x)) | X] \\ &= E[w(\theta)(g(\theta) - \delta(x))^2 | X] \\ &= E[w(\theta)(g(\theta)^2 - 2g(\theta)\delta(x) - \delta(x)^2) | X] \\ &= E[w(\theta)g(\theta)^2 | X] - 2\delta(x)E[w(\theta)g(\theta) | X] + \delta(x)^2E[w(\theta) | X]. \end{aligned}$$

To find the Bayes estimator, we take the derivative with respect to $\delta(x)$,

$$\frac{\partial \rho(\theta, \delta(x))}{\partial \delta(x)} = -2E[w(\theta)g(\theta) | X] + 2\delta(x)E[w(\theta) | X]$$

and solve for $\hat{\delta}(x)$ when $\frac{\partial \rho(\theta, \delta(x))}{\partial \delta(x)} = 0$. Observe

$$0 = -2E[w(\theta)g(\theta) | X] + 2\hat{\delta}(x)E[w(\theta) | X].$$

Solving for $\hat{\delta}(x)$ gives $\hat{\delta}(x) = \frac{E[w(\theta)g(\theta) | X]}{E[w(\theta) | X]}$. To check if the solution is unique, it must be true that $\frac{\partial^2 \rho(\theta, \delta(x))}{\partial \delta(x)^2} > 0$. Here, we find

$$\frac{\partial^2 \rho(\theta, \delta(x))}{\partial \delta(x)^2} = 2 * E[w(\theta) | X] > 0.$$

since we assume $w(\theta) > 0$. Thus, our Bayes estimator is unique. QED